Conference Paper Title*

*Note: Sub-titles are not captured in Xplore and should not be used

1st Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

2nd Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

3rd Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

4th Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

5th Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

6th Given Name Surname
dept. name of organization (of Aff.)
name of organization (of Aff.)
City, Country
email address or ORCID

***Abstract*—Since photorealistic faces can be readily generated by facial manipulation technologies nowadays, potential malicious abuse of these technologies has drawn great concerns and societal risks. This emphasizes the need for more reliable detection techniques that can represent temporal and spatial inconsistencies in video data. Deep learning (DL) methods have become effective instruments for examining and spotting minute discrepancies that set real information apart from deepfakes. In this paper, we propose a spatiotemporal deepfake video detection framework that combines a Bidirectional Long Short-Term Memory (BiLSTM) network for temporal sequence modeling with a pre-trained ResNet50 Convolutional Neural Network (CNN) for spatial feature extraction. The ResNet50 backbone recovers high-level visual representations from individual face frames, capturing minute texture imperfections and spatial inconsistencies, while the BiLSTM models temporal relationships across subsequent frames to detect frame-level anomalies typical of deepfake movies. Face areas are automatically extracted by a YOLOv8-based face detection module, enabling dependable preprocessing in a range of video scenarios.**

The proposed methodology achieves a validation accuracy of 88.89% and a test-set accuracy of 86.00% with strong precision, recall, and F1-score performance, according to experimental results. These results show that holistically modeling spatial and temporal information greatly enhances the durability of deepfake detection as compared to image-based methods. This study improves the deepfake detection algorithm and shows how it may be used practically in media integrity and digital forensics.

***Index Terms*—Deepfake Detection Spatiotemporal Analysis ResNet50 Bidirectional LSTM Video Forensics Deep Learning**

I. Introduction

With recent advances in deep generative models, increasing hyper-realistic face images or videos can be readily generated, which can easily cheat human eyes. This leads to serious misinformation and fabrication problems in politics, entertainment and society. Neural networks, once these techniques are maliciously abused. This threat is known as DeepFake. Generative adversarial networks (GANs), are the foundation of deepfake technology because they allow for the smooth imitation of human speech patterns and features. This makes it possible to produce videos in which people seem to say or do things they haven't done. Deepfake technology presents ethical and societal issues even though it offers enormous creative and educational potential in fields like virtual reality, gaming, filmmaking, and language learning. Its inappropriate use, such as disseminating false information, generating sexual content without authorization, or impersonating others, endangers privacy, security, and confidence. Rapid technology growth needs the development of effective detection tools and moral standards in order to prevent abuse and maximize its positive potential. The rapid growth of deepfake technology has transformed video creation by allowing AI powered ways to create simulated media that is extremely lifelike. But there are still a lot of obstacles to overcome, especially when it comes to stopping the malicious use of deepfakes for identity theft, disinformation, and privacy violations. Deepfake detection has new uses in IoT-enabled and smart systems, going beyond stand-alone media forensics. Deepfake detection models can be used in smart cities to authenticate video streams used for law enforcement, traffic monitoring, and public safety by integrating them with surveillance networks, smart cameras, and edge devices. By stopping identity spoofing through manipulated video feeds, deepfake detection can improve biometric authentication in smart homes and access control systems.Using edge based inferences techniques media monitoring systems like cameras and sensors spread across many locations can use deepfake detection directly on local devices .This method is energy efficient and effective.The urgent need for more reliable and intelligent detection systems is highlighted by the fact that deepfake generation tools are frequently becoming more sophisticated than traditional detection techniques. The technical contributions of the paper are as

Identify applicable funding agency here. If none, delete this.

follows :A spatiotemporal deepfake video detection framework is put forth, which combines a Bidirectional LSTM to detect temporal inconsistencies between video frames with a pre-trained ResNet50 CNN for spatial feature extraction. To accomplish reliable and automated face localization in a variety of pose, lighting, and background scenarios, a YOLOv8-based face detection module is integrated. A balanced deepfake video dataset is used for extensive testing, and metrics such as accuracy, precision, recall, and F1-score are used to assess performance. The impact of temporal modeling is examined through an ablation study, which shows that bidirectional sequence learning is more effective than spatial-only and unidirectional baselines.

II. Literature Survey

As researchers use a combination of spatial, temporal, and even physiological strategies to handle ever more clever manipulation tricks, deepfake detection has rapidly advanced. Newer research tends to combine spatiotemporal patterns with attention mechanisms for better outcomes, whereas earlier attempts focused just on spatial aspects. Shang et al.'s Pixel-Region Relation Network (PRRNet), which uses both region-wide patterns and pixel-level information to identify face forgeries [25]. It establishes new standards on a number of important datasets by identifying small local defects. Wang et al. shifted focus to noise patterns, using a Siamese noise extractor that splits faces from backgrounds to reveal mismatches in deepfake images [1]. It offers a clearer perspective on forensics media detection. Texture analysis has proven equally powerful. Yang et al.'s forensic setup zeroes in on faint texture mismatches in real versus fake faces, delivering top accuracy and underscoring cues our eyes often miss [5]. Nawaz et al. pushed architecture innovation with ResNet-Swish-Dense54, merging residuals, Swish activations, and dense links for robust deepfake detection [2]. They back it up with explainable heatmaps and tests against datasets with a lot of tweaks. Attention mechanisms are popping up everywhere lately. Wang et al.'s Attention-expanded Deepfake Spotter (ADS) uses region maps and smart weighting to boost cross-dataset reliability [4]. Waseem et al. took it further with multi-attention for pinpointing face swaps and expression edits [5].

Muruganandham, P, et al.'s presented a deepfake audio detection LSTM-based auto-encoder with Dynamic Residual Difference Encoding, which combines residual-based feature discrimination with attention-enhanced temporal modeling. [29]

Chunlei Pen et al. propose DeepFidelity, a quality-aware Deepfake detection framework that models perceptual forgery fidelity to distinguish real and fake faces across varying image qualities. Their method incorporates a symmetry-based Vision Transformer (SSAAFormer) to enhance spatial feature modeling and demonstrates superior performance on multiple benchmark datasets. [**?**]

Lately, biological signals are turning heads too. Xu et al. tapped remote photoplethysmography (rPPG) to catch fake videos via irregular heart-rate signals [6]. Usmani et al. kept it lightweight with a shallow vision transformer that mixes attention for solid results without heavy compute [7].

III. Datasets Used For The Model

A range of large-scale datasets are used to guarantee diversity in manipulation methods, ambient circumstances, and subjects for reliable training, particularly when addressing low- quality videos, evaluation, and benchmarking of deepfake detection systems. One of the most well-known datasets, FaceForensics++, includes both original and manipulated videos generated using different compression settings and face-editing techniques. It provides controlled recording conditions, which makes it suitable for supervised training and performance evaluation specific to manipulation artifacts. Facebook released the Deepfake Detection Challenge (DFDC) dataset, which consists of thousands of manipulated videos produced using multiple deepfake generation pipelines under various real-world conditions, including visual distortions, diverse backgrounds, and lighting variations. As this dataset was designed to represent challenging scenarios encountered on social media platforms, it is particularly important for assessing real-world robustness and generalization capabilities.

By providing over 1000 real and manipulated videos with diverse factor,like head poses, facial expressions, and lighting conditions, the DeepForensics dataset focuses on facial manipulation detection. It is well suited for bias analysis and realistic manipulation detection, as it includes deepfakes generated using advanced algorithms, along with participants of varying ages, genders, and ethnicities. DeeperForensics1.0 further improves realism by incorporating 48,475 authentic videos and 11,000 manipulated videos created using an end-to-end face-swapping model (DF-VAE). It enables evaluation under noisy and degraded visual conditions by introducing several real-world perturbations of varying intensities, such as motion blur, noise, and compression artifacts. Finally, Celeb-DF represents a significant improvement in realism over earlier datasets, containing 5,639 high-quality deepfake videos and 590 authentic celebrity videos with minimal artifacts. It evaluates models' ability to detect subtle indicators of manipulation and remains a reliable benchmark for assessing performance under visually convincing conditions.

Proposed Methodology

A. Process Flow

The creation of deepfake videos introduces artifacts that appear in both temporal and spatial dimensions. Deep convolutional feature hierarchies are an efficient way to capture the high-frequency texture inconsistencies, blending artifacts, and localized facial distortions that arise from frame-level manipulations. Sequential modeling is necessary to reliably detect temporal inconsistencies in motion dynamics, facial expressions, and inter-frame coherence that are frequently produced by deepfake synthesis. Using a convolutional neural network to extract discriminative spatial embeddings from individual frames and a Bidirectional Long Short-Term Memory network

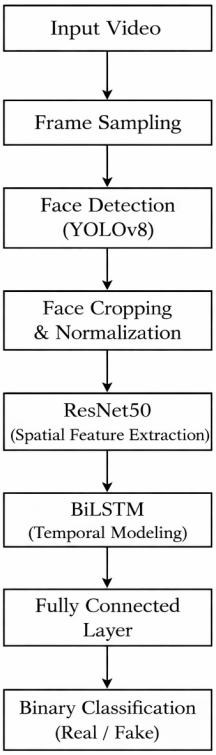


Fig. 1: Block diagram of the proposed ResNet50–BiLSTM based deepfake video detection framework.

to model contextual relationships and long-range temporal dependencies across frame sequences, the proposed architecture addresses these features. By utilizing complementary spatial and temporal cues, this spatiotemporal decomposition allows for reliable video-level deepfake detection. The overall process flow of the proposed deepfake detection system is illustrated in Fig. 1 and consists of all the steps performed during the making of the model.

In the first stage of this model a video is acquired for analysis. The system is designed to process videos of varying lengths, resolutions, and quality levels. Handling diverse video with varying factors is essential to ensure robustness in real-world scenarios, where input media may be affected by compression artifacts, motion blur, or low resolution. Such inconsistencies help increase the reliability of a deepfake detection model

In the second stage, the input video is decomposed into individual frames . Frames are extracted at a fixed sampling interval to reduce redundancy while preserving temporal continuity across consecutive frames.

YOLOv8-based face detection model is used for face detection on the sampled frames it is an efficient and robust approach for localization of facial regions across frames under different poses, illumination conditions, and background complexity. thereby improving training stability and feature consistency.

To maintain contextual facial information, detected face regions are chopped from the original frames with an extra spatial margin. They are then enlarged to a fixed resolution of 224×224 pixels to meet the convolutional neural network's input requirements. In order to guarantee uniform input distributions and enhance training stability and feature consistency, pixel values are normalized.

ResNet50, a deep convolutional neural network pre-trained on the ImageNet dataset, is used to extract spatial features. With its 50 layers arranged into residual blocks with skip connections, ResNet50 successfully addresses the vanishing gradient issue and makes it possible to learn deep hierarchical representations. By using pre-trained ImageNet weights, transfer learning enables the network to extract rich and discriminative visual information without the need for new training.

After the final fully connected layer of ResNet50 is removed for feature extraction, each input face image has a 2048-dimensional feature vector. While the early convolutional layers are locked during training to balance generalization and task-specific learning, the last residual block is adjusted to adapt high-level representations to deepfake-specific artifacts.

For temporal modeling, a Bidirectional Long Short-Term Memory (BiLSTM) network is employed, which examines feature vector sequences extracted from subsequent frames. The BiLSTM design incorporates both forward and backward temporal passes, allowing the model to simultaneously learn contextual associations from past and future frames. This bidirectional modeling is particularly effective at identifying the subtle temporal anomalies that are characteristic of deepfake films.

For each video,30 feature vectors, are provided as input to the BiLSTM network. The BiLSTM comprises of 256 hidden units in each direction, resulting in a 512-dimensional representation after temporal aggregation. Dropout regularization with a rate of 0.5 is applied in the classification stage to reduce overfitting and improve generalization performance.

The aggregated spatiotemporal representation produced by the BiLSTM is forwarded to the final classification layer,which determines whether the input video is authentic or manipulated.

B. Implementation Details

The implementation employs a custom dataset loader, *VideoDataset*, designed to handle video-based deepfake data.

TABLE I: Dataset statistics

Description	**Count / Value**
Total video folders	3000
Real video folders	1500
Fake video folders	1500
Frames per video folder	30
Frame resolution	224×224
Class balance (Real:Fake)	1:1

Data Split: The dataset is divided into training, validation, and test splits by keeping a balanced class balance

- Training set: 2,100 video folders (70%)
- Validation set: 450 video folders (15%)
- Test set: 450 video folders (15%)

Training Configuration:

- **Optimizer:** AdamW optimizer with a learning rate of 3×10^{-4}
- **Batch Size:** 4 video clips per batch
- **Epochs:** 12
- **Device:** CUDA-enabled GPU
- **Loss Function:** Binary Cross-Entropy with Logits loss
- **Evaluation Metrics:** Accuracy, Precision, Recall, and F1-score

Input: Sequences of 30 extracted face frames resized to 224×224

Output: Video-level deepfake probability score in the range $[0, 1]$

C. Model Components

- **CNN Backbone:** ResNet50 (pretrained, partially fine-tuned)
  - Pretrained on the ImageNet dataset
  - Final fully connected layer removed
  - Output: 2048-dimensional feature vector per frame
  - Early convolutional layers frozen, with fine-tuning applied to the final residual block
- **BiLSTM:** Bidirectional Long Short-Term Memory network
  - Input size: 2048
  - Hidden size: 256 per direction
  - Number of layers: 1
  - Bidirectional: forward and backward temporal modeling
- **Classifier Head:** Fully connected classification module
  - Linear layer: $512 \rightarrow 256$
  - ReLU activation
  - Dropout with rate 0.5
  - Linear layer: $256 \rightarrow 1$

D. Training Results

The performance of the proposed model during the training loop is summarized in Table II. The results report training loss, validation loss, and validation accuracy across twelve representative epochs, illustrating models learning behavior and convergence.

The model achieved its highest accuracy of **88.89%** at the ninth epoch, indicating effective convergence and strong generalization capability.

A confusion matrix based on the test set is computed in Fig. 2. The matrix highlights true positives and false negatives to show how well the model can differentiate between actual and bogus video samples.

TABLE II: Training and validation performance across epochs

Epoch	Train Loss	Val Loss	Val Accuracy
1	0.5697	0.3961	0.8467
2	0.4639	0.3533	0.8444
3	0.3842	0.3415	0.8422
4	0.3380	0.3831	0.8444
5	0.2563	0.3167	0.8644
6	0.2092	0.4862	0.8200
7	0.1838	0.4442	0.8378
8	0.1524	0.3198	0.8667
9	0.1193	0.3679	**0.8889**
10	0.1001	0.5530	0.8689
11	0.1112	0.4346	0.8400
12	0.0825	0.6737	0.8133

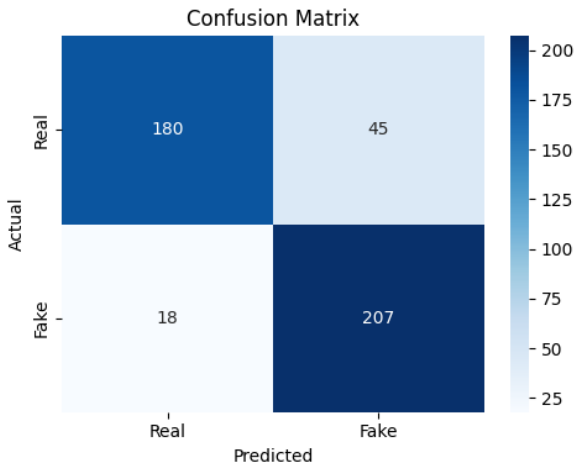


Fig. 2: Confusion matrix evaluated on the test set.

IV. Results and Discussion

The proposed architecture illustrates stable performance for deepfake video detection by effectively leveraging both spatial and temporal information. A summary of the key quantitative results obtained during training, validation, and final evaluation of the experiment is presented in Table III.

The training process shows steady improvement performance across the epochs .The highest validation accuracy of 88.89% was achieved at the ninth epoch, after which validation performance fluctuated while training loss continued to decrease, suggesting the onset of mild overfitting in later epochs.Early stopping based on validation loss was used during training to prevent overfitting; the optimization process was stopped when no improvement was seen over subsequent epochs. In order to enhance generalization and lower model variance, dropout regularization was also implemented in the classification head at a rate of 0.5.The increase in the validation loss after 9th epoch motivates the use of early stopping in the training pipeline

To assess the models' capability to on unseen data the final evaluation was conducted on test set. The model achieved an accuracy of 86.00%, with a precision of 82.14%, recall of 92.00%, and an F1 score of 86.79%, demonstrating a a low false-negative rate. The corresponding ROC and Precision–

TABLE III: Summary of training, validation, and test performance

Metric	Value
Best validation accuracy (Epoch 9)	88.89%
Final training loss (Epoch 12)	0.0825
Final validation loss (Epoch 12)	0.6737
Test set accuracy	86.00%
Test set precision	82.14%
Test set recall	92.00%
Test set F1-score	86.79%

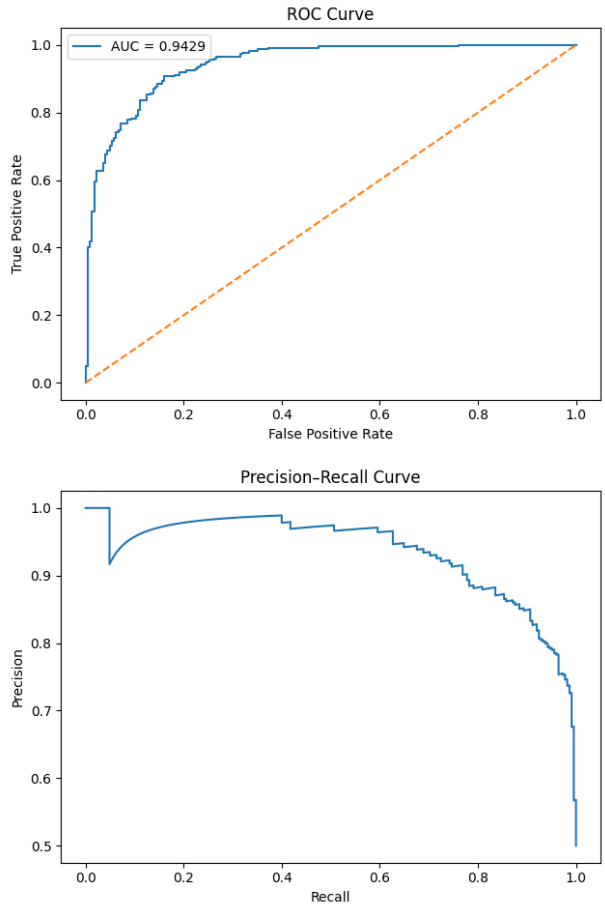


Fig. 3: ROC and Precision–Recall curves evaluated on the test set.

Recall curves evaluated on the test set are shown in Fig. 3, highlighting the discriminative capability of the proposed approach across a range of decision thresholds.

A. Ablation study

An ablation study was conducted on three architectural variants: a CNN-only model, a CNN combined with a unidirectional LSTM, and the proposed CNN–BiLSTM architecture. All models were trained and evaluated under identical experimental settings and same dataset, and the results are summarized in Table 4.

The results prove that spatial analysis alone is not sufficient to make a optimal model hence adding temporal modeling improves the performance. Furthermore, the bidirectional LSTM uses leveraging contextual information from both past and future frames outperforming unidirectional LSTM.

TABLE IV: Ablation study across different models

Model	Accuracy	Precision	Recall	F1-score
CNN-only	63.“%	64.39%	58.67%	61.40%
CNN + LSTM	81.00%	83.00%	81.00%	81.00%
CNN + BiLSTM	86.00%	82.14%	92.00%	86.79%

Using transfer learning with a pre-trained ResNet50 backbone decreases training time and compute needs while enhancing feature representation quality. By freezing early convolutional layers and fine-tuning the residual block, the model preserves generic visual features while adjusting high-level representations to deepfake-specific artifacts. This improves generalization on restricted datasets.

Overall, the the proposed system ensures scalability and adaptability. The framework can be extended to support longer video sequences, larger datasets with varying factors like blurriness and lighting .

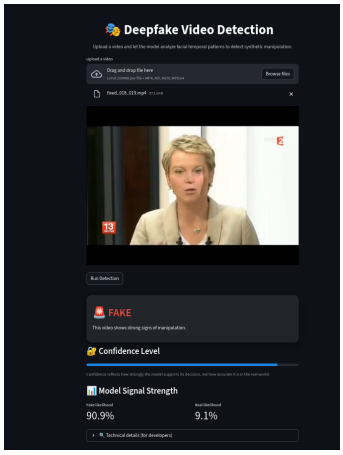
Thus, under the same experimental conditions, the ablation analysis acts as an internal benchmark, offering a controlled and equitable comparison between spatial-only and spatiotemporal architectures. Future research will concentrate on expanding this evaluation to standardized deepfake benchmarks and cross-dataset scenarios in order to further assess robustness and generalization across a variety of real-world conditions, even though the suggested CNN–BiLSTM framework consistently outperforms CNN-only and CNN+LSTM baselines.

B. Qualitative Results

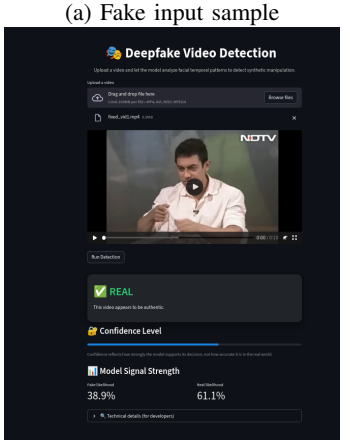
Fig. 4 presents the example of output provided by the model on both real and manipulated inputs. The results indicate the models accuracy with the confidence correctly classifying videos as real and fake

V. Conclusion

This work investigates the high-level semantics for deepfake detection. The study provides a robust framework that includes a Bidirectional Long Short-Term Memory (BiLSTM) network for temporal inconsistencies across video frames, a YOLOv8-based face detection module for precise face localization, and ResNet50 for high-quality spatial feature extraction. The proposed architecture outperforms standard image-based deepfake detection algorithms, including CNN and CNN+LSTM models, with a validation accuracy of 88.89%. In order to increase generalization against rapidly changing deepfake generation methodologies, future research will also concentrate on refining the framework for real-time deployment and carrying out thorough cross-dataset evaluations. The model can be further used to detect real time deepfake detection using quantization techniques such as INT8 quantization to reduce model size and memory this can further be evaluated using metrics such as frames per second , end to end latency ,and energy consumption on edge and GPU-based platforms.These



(a) Fake input sample



(b) Real input sample

Fig. 4: Qualitative results showing model predictions for fake and real inputs.

enhancements will enable scalable and sustainable deployment of the system in latency-sensitive and resource-constrained environments. Overall, our work supports the creation of dependable and trustworthy digital media verification systems and offers a strong basis for deepfake forensics advancement.

References

[1] Wang, X., Zhang, Y., Li, H.: Siamese noise extractor with face-background strategy for deepfake detection. *IEEE Trans. Inf. Forensics Security* (2020) https://www.sciencedirect.com/science/article/pii/S2666281722000762

[2] Nawaz, M., Irfan, M., Khan, S.: ResNet-swish-dense54 for robust deepfake detection and explainability. *Computers & Security* (2022) https://doi.org/10.1007/s00371-022-02732-7

[3] Ismail, A., Hassan, R., Ali, F.: Integrated spatiotemporal deepfake detection using YOLO and hybrid network. *Neural Computing and Applications* (2022) https://link.springer.com/article/10.1007/s00521-022-07633-3

[4] Wang, T., Liu, X., Chen, Z.: Attention-expanded deepfake spotter: A two-stage forgery detector. *IEEE Access* (2023) https://doi.org/10.1007/s11042-023-17672-0?urlappend=%3Futm_source%3Dresearchgate.net%26utm_medium%3Darticle

[5] Waseem, H., Rehman, A., Qureshi, M.: Multi-attention deepfake detection and localization. *Expert Systems with Applications* (2023) https://arxiv.org/abs/2103.02406

[6] Xu, L., Zhang, C., Li, J.: Deepfake detection using remote photoplethysmography signals. *IEEE Trans. Biometrics, Behavior, and Identity Science* (2023) https://ieeexplore.ieee.org/document/11214405

[7] Usmani, M., Khan, R., Ahmed, Z.: Shallow vision transformer for efficient deepfake detection. *Applied Intelligence* (2024) https://link.springer.com/article/10.1007/s11042-023-15910-z

[8] Yang, L., Chen, F., Wang, J.: Texture-based intelligence forensic method for deepfake detection. *Pattern Recognition Letters* (2021) https://www.sciencedirect.com/science/article/abs/pii/S0167739X21002387

[9] Afchar, D., Nozick, V., Yamagishi, J., Echizen, I.: MesoNet: A compact facial video forgery detection network. *arXiv:1809.00888* (2018) https://ieeexplore.ieee.org/document/8630761

[10] Rössler, A. et al.: FaceForensics++: Learning to detect manipulated facial images. In: *Proc. IEEE/CVF ICCV* (2019) https://arxiv.org/abs/1901.08971

[11] Li, Y., Lyu, S.: Exposing deepfake videos by detecting face warping artifacts. In: *Proc. CVPR Workshops* (2019) https://arxiv.org/abs/1811.00656

[12] Haliassos, A. et al.: Lips don't lie: A generalizable and robust approach to face forgery detection. In: *Proc. CVPR* (2021) https://arxiv.org/abs/2012.07657

[13] Durall, R. et al.: Watch your up-convolution: CNN-based generative deep neural networks are failing to reproduce spectral distributions. In: *Proc. CVPR* (2020) https://arxiv.org/abs/1912.13458

[14] Li, L. et al.: Face X-ray for more general face forgery detection. *arXiv:1912.13458* (2019) https://arxiv.org/abs/1912.13458

[15] Qian, Y. et al.: Thinking in frequency: Face forgery detection by mining frequency-aware clues (F3Net). In: *Proc. ECCV* (2020) https://arxiv.org/abs/2007.09355

[16] Yu, N., Davis, L.S., Fritz, M.: Attributing fake images to GANs: Learning and analyzing GAN fingerprints. In: *Proc. ICCV* (2019) https://arxiv.org/abs/1811.08180

[17] Nguyen, H.H., Yamagishi, J., Echizen, I.: Use of a capsule network to detect fake images and videos. *arXiv:1910.12467* (2019) https://arxiv.org/abs/1910.12467

[18] Ismail, A. et al.: An integrated spatiotemporal-based methodology for deepfake detection. *Neural Computing and Applications* (2022) https://link.springer.com/article/10.1007/s00521-022-07633-3

[19] Wang, T., Chow, K.-P.: Deepfake noise investigation and detection. *Forensic Science International: Digital Investigation* (2022) https://doi.org/10.1016/j.fsidi.2022.301395

[20] Ciftci, U.A., Demir, I., Yin, L.: FakeCatcher: Detection of synthetic portrait videos using biological signals. *arXiv:1901.02212* (2019) https://ieeexplore.ieee.org/document/9141516/citations#citations

[21] Qi, H. et al.: DeepRhythm: Exposing deepfakes with attentional visual heartbeat rhythms. In: *Proc. ACM MM* (2020) https://arxiv.org/abs/2006.07634

[22] Hernandez-Ortega, J. et al.: DeepfakesOn-Phys: Deepfake detection based on heart rate estimation. *arXiv:2010.00400* (2020) https://arxiv.org/abs/2010.00400

[23] Yang, J. et al.: Detecting fake images by identifying potential texture difference. *Future Generation Computer Systems* (2021) https://doi.org/10.1016/j.future.2021.06.043

[24] Kingra, S. et al.: LBPNet: Exploiting texture descriptor for deepfake detection. *Forensic Science International: Digital Investigation* (2022) https://doi.org/10.1016/j.fsidi.2022.301452

[25] Zhao, H. et al.: Multi-attentional deepfake detection. In: *Proc. CVPR* (2021) https://arxiv.org/abs/2103.02406

[26] Ilyas, H. et al.: AVFakeNet: A unified end-to-end dense Swin transformer for audio-visual deepfake detection. *Applied Soft Computing* (2023) https://doi.org/10.1016/j.asoc.2023.110124

[27] Shang, Z., Xie, H., Zha, Z., Yu, L., Li, Y., Zhang, Y.: PRRNet: Pixel-region relation network for face forgery detection. In: *Proc. IEEE/CVF CVPR* (2021) https://doi.org/10.1016/j.patcog.2021.107950

[28] Chunlei Peng, Huiqing Guo, Decheng Liu, Nannan Wang, Ruimin Hu, Xinbo Gao.: DeepFidelity: Perceptual Forgery Fidelity Assessment for Deepfake Detection https://doi.org/10.48550/arXiv.2312.04961

[29] Muruganandham, P., Thangasamy, G.R., Jayaraman, S.: . LSTM autoencoder based parallel architecture for deepfake audio detection with dynamic residual encoding and feature fusion. https://doi.org/10.1038/s41598-025-08198-6